

Alexander Short

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EDUCATION

Queen's University

Sept. 2021 – Apr. 2025

Bachelor of Applied Science in Electrical Engineering (BASc), Highest Honours

University of Toronto

Sept. 2025 – Expected Dec. 2026

Master of Engineering in ECE (MEng)

Relevant Coursework: Embedded Systems, Computer Security, Digital Signal Processing, Integrated Circuits, Microwave Circuits, Power Electronics

WORK EXPERIENCE

Laszlo Energy Services | *Electronics Engineer Intern*

Summers 2023–2025; part-time 2023–2024

- Designed production PCBs for a hybrid-heating HVAC control system, including distributed temperature-sensing and controller hardware.
- Developed embedded communication and over-the-air firmware-update functionality for remote sensor boards.
- Supported PCB fabrication, assembly, validation, and deployment of operational systems across multiple customer homes, including custom 3D-printed enclosures.

PROJECTS

Embedded Multiprotocol Smart Home Gateway | *Zigbee, Thread, Embedded Linux, KiCad* June 2025 – Present

- Designed a custom 8-layer high-density embedded Linux PCB integrating a 0.5 mm-pitch BGA AM6231 SoC, DDR4, Gigabit Ethernet, eMMC, USB, power management, and a Silicon Labs MGM240P radio.
- Brought up custom hardware from initial power-on through U-Boot and Linux, debugging power-rail, sequencing, Ethernet PHY, and USB issues across two PCB revisions.
- Rebuilt and configured the Linux kernel and device tree to support custom hardware and multicast networking.
- Integrated OpenThread Border Router and Silicon Labs multiprotocol Zigbee/Thread software on the AM6231, communicating with the MGM240P RCP over CPC/UART.
- Commissioned and controlled commercial Zigbee and Thread/Matter devices, validating routing, reconnection, and persistent operation across system power cycles.

Custom FPV Flight Controller | *STM32N6, KiCad, RF / High-Speed PCB Design*

Aug. 2026 – Present

- Designing a 6-layer mixed-signal/RF flight controller PCB around an STM32N6 MCU, integrating MIPI CSI video, 5 GHz wireless, 915 MHz control, high-speed memory, navigation sensors, and ESC interfaces.
- Designed RF signal chains with controlled-impedance CPWG routing, balun/filter networks, antenna matching, RF keepouts, and VNA-accessible tuning networks
- Architected a 6S LiPo power system with multi-rail regulation, USB-C/battery power-path management, protection and monitoring circuitry
- Developing bare-metal/FreeRTOS STM32N6 firmware for sensor acquisition, flight-control interfaces, and wireless communication, integrating SPI/I²C/UART peripherals and real-time interrupt-driven processing.

Networked LED Matrix Canvas in Embedded Rust | *Rust, ESP32*

Sept. 2025 – Dec. 2025

- Developed `no_std` embedded Rust firmware for an ESP32-C6 to retrieve and display a remotely controlled 16×32 pixel canvas on an LED matrix.
- Implemented Wi-Fi station connectivity, DHCP, and a blocking TCP/IP stack using `esp-hal`, `esp-radio`, and `smoltcp`.
- Built an HTTP/JSON data pipeline using `serde_json_core` and fixed-capacity `heapless` structures, managing socket lifetimes, network buffers, and errors without dynamic allocation.

SKILLS

Programming: C, Rust, Python, MATLAB, VHDL

Embedded: Embedded Linux, U-Boot, Device Trees, OpenThread, Zigbee, Matter

Interfaces & Networking: UART, SPI, I²C, GPIO, USB, Ethernet, TCP/IP

Hardware & EDA: KiCad, Cadence Virtuoso, Keysight ADS, LTspice, PSim, Quartus Prime, FPGA, Soldering

CAD: SolidWorks, Onshape